Wenhan LUO, Ph.D., IEEE Senior Member

Contact Information

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Working Experience

The Hong Kong University of Science and Technology — **Hong Kong**
- Associate Professor — Jun, 2024 – present

Sun Yat-sen University — **Shenzhen, China**
- Associate Professor — Jul, 2022 – Mar, 2024

Tencent — **Shenzhen, China**
- Senior Researcher — Mar, 2020 – Jul, 2022

Amazon — **Palo Alto, CA, USA**
- Research Scientist — Jul, 2019 – Jan, 2020

Tencent — **Shenzhen, China**
- Senior Researcher — Aug, 2016 – May, 2019

Education

Imperial College London — **London, UK**
- Ph.D. in Computer Science — Oct, 2012 – Jun, 2016

Institute of Automation, Chinese Academy of Sciences — **Beijing, China**
- M.E. in Control Theory and Control Engineering — Sep, 2009 – Jul, 2012

Huazhong University of Science and Technology — **Wuhan, China**
- B.E. in Automation — Sep, 2005 – Jul, 2009

Research Interests

Creative AI, Generative Model, Computer Vision, Machine Learning

Research Output Statistics

- Citations - Google Scholar: 16146 (H-index: 59)
- 90+ papers at the best conferences and journals in computer vision and machine learning, including IEEE TPAMI (9), IJCV (7), Nature Communications (1), IEEE TIP (10), CVPR (23), ICCV (8), ECCV (11), NeurIPS (9), ICML (4), ICLR (6), SIGGRAPH (2), AAAI (3), ACL (2), ACM MM (3), Artificial Intelligence (1), IEEE TVCG (1), IEEE TKDE (2)
- 10+ papers are recognized as Highly Cited Papers or Hot Papers by Clarivate

Awards and Honors

- IEEE Signal Processing Society Outstanding Editorial Board Member Award, 2026.
- AAAI Outstanding Senior Program Committee Service Award, 2026
- CCF-Tencent Rhino-Bird Faculty Fund Excellence Award, 2024
- Champion award in the Bracketing Image Restoration and Enhancement Challenge - Track 1 & 2, NTIRE 2024, in conjunction with CVPR 2024
- Second place in the Few-shot RAW Image Denoising competition of MIPI 2024, in conjunction with CVPR 2024
- CCF-Tencent Rhino-Bird Fellowship, 2023.
- Top 2% Scientists Worldwide, 2023, 2024, 2025 & 2026.
- ACM China Rising Star (Guangzhou Chapter), 2022.
- CVPR Best Paper Finalist, 2019.

Research Grants

Grants as Principal Investigator
- Towards Efficient Sparse Multimodal Large Language Models for Holistic Modality Understanding, China National Postdoctoral Council under the Ministry of Human

Resources and Social Security, 2027/01 - 2029/01
- Large Model for Multimodal Understanding, Gift Fund, Baidu, 2026/10 - 2027/09
- Research on Generalizable Visual Restoration for Open-World Scenarios, AIS Support Fund for Interdisciplinary Research Collaboration, 2026/09 - 2027/08
- Towards Efficient, Adaptive, and Scalable Visual Generative Models for Edge Devices, General Research Fund, Research Grants Council of Hong Kong, 2026/07 - 2029/06
- AIGC video generation/editing and world models in film and television, Gift Fund, Tencent, 2026/07 - 2027/06
- Controllable High-Precision 3D Scene Generation Based on Spatial Relationship Awareness and Multimodal Guidance, Gift Fund, Tencent, 2025/12 - 2026/12
- Efficient Training Methods for Image Generation Models, Gift Fund, Kuaishou, 2025/12 - 2026/12
- Coherent Multi-subject Video Generation, Contract Research, ByteDance, 2025/09 - 2026/09
- AI Digital Human Research, Gift Fund, Video Rebirth Limited, 2025/07 - 2029/07
- New Architecture of Large Model, Gift Fund, Hong Kong Wiener Intelligence Technologies Limited, 2025/01 - 2026/12
- Research on Large-scale Fine-tuning and Alignment of Mixture-of-expert Large Models, HKUST-MetaX Joint Lab Fund, 2024/11 - 2026/09
- Research on Key Algorithms of Face Restoration, Perception and Understanding for Under-display Cameras, General Program, National Natural Science Foundation of China, 2024/01 - 2027/12
- Efficient Video Restoration, Contract Research, Samsung, 2023/07 - 2024/07
- Towards Cross-domain Matching: Research on Multi-path Model for Advertisement Relevance, Contract Research, Tencent, 2023/06 - 2024/09
- Research on Image and Video Restoration in Autonomous Driving Scenarios under Adverse Weather, General Program, Guangdong Basic and Applied Basic Research Foundation, 2023/01 - 2025/12

Grants as Co-PI
- Solid-Liquid Biphase Spatial Domain Ultrasonic Ultra-Low Water Consumption Plant-Based Fiber Mechanical Extraction Mechanism - AgriFiber Pro: From Agricultural Waste to Product, Platform Research Programme, Hong Kong Innovation and Technology Commission, 2025/12 - 2027/12
- Research and Development of Key Technologies for Multimodal Recognition and Generative Artificial Intelligence under Large Models, as well as Development and Industrialization of Intelligent Sports Scenarios, International Cooperation Program, Qingdao Municipal Bureau of Science & Technology, 2025/04 - 2027/04
- Research and Development of Key Technologies for Intelligent Generation of High-Quality Video Covers of Massive Video, Technology Development Key Project, Shenzhen Science and Technology Innovation Commission, 2022/11 - 2025/10

Selected Publications

For the complete list, please see Google Scholar

1. S. Chen, A. Liu, K. Di, Y. Xu, H. Ye, **W. Luo**, N. Zou, FedDyMem: Efficient Federated Learning with Dynamic Memory and Memory-Reduce for Unsupervised Image Anomaly Detection, *International Journal of Computer Vision (**IJCV**)*, to appear.
2. Z. Lu, T. Liu, J. Huang, W. Zhao, X. Liu, **W. Luo**, M. Wang, FLAME: Flow Guidance and Adaptive Frequency Fusion for All-in-One Mamba-Enhanced Video Restoration, *IEEE Trans. on Image Processing (**TIP**)*, to appear.
3. Z. He, M. Yang, S. Yang, Z. Lai, X. Yue, C. Guo, **W. Luo**, MaterialMVP++: High-quality Material Generation with Neural Bump Estimation via Multi-view PBR

Diffusion, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, to appear.

4. Z. He, M. Yang, S. Yang, H. Sun, X. Han, C. Guo, **W. Luo**, Tango3D: Towards Alignment for Global and Local 2D-3D Correspondence, *Neural Information Processing Systems (**NeurIPS**)*, 2026.
5. Y. Cai, N. Chen, R. Xie, J. Pan, C. Jiang, C. Chen, W. Zhou, Z. Sun, W. Xue, **W. Luo**, Y. Guo, HOMIE: Human-object Centric Video Personalization via Multimodal Intelligent Enhancement, *Neural Information Processing Systems (**NeurIPS**)*, 2026.
6. Y. Weng, **W. Luo**, Q. Shao, VCR: Learning Valid Contextual Representation for Incomplete Wearable Signals, *Neural Information Processing Systems (**NeurIPS**)*, 2026.
7. B. Tang, T. Wang, X. Yu, **W. Luo**, K. Zhang, B. Li, M. Zhang, World-Model-Inspired Flicker State Modeling for Burst Flicker Removal, *Neural Information Processing Systems (**NeurIPS**)*, 2026.
8. H. Li, J. Wang, S. Cai, H. Sheng, B. Deng, J. Ye, **W. Luo**, EchoStyle: Unlocking High-Fidelity Video Stylization with Reverse Data Synthesis, *European Conference on Computer Vision (**ECCV**)*, 2026.
9. H. Sun, M. Yang, S. Yang, Z. He, X. Han, H. Fu, C. Guo, **W. Luo**, ROAR-3D: Routing Arbitrary Views for High-Fidelity 3D Generation, *European Conference on Computer Vision (**ECCV**)*, 2026.
10. P. Li, R. Khirodkar, J. Li, Y. Dong, C. Cao, Y. Liu, **W. Luo**, Y. Guo, S. Saito, LUNA: Learning Universal 3D Human Animation Beyond Skinning, *European Conference on Computer Vision (**ECCV**)*, 2026.
11. Y. Jiang, T. Wang, Z. Wang, C. Cao, J. Wu, **W. Luo**, W. Hu, J. Gao, C. Guo, *European Conference on Computer Vision (**ECCV**)*, 2026.
12. J. Deng, T. Weng, T. Yang, **W. Luo**, Z. Li, W. Jiang, LlamaSeg: Image Segmentation via Autoregressive Mask Generation, *European Conference on Computer Vision (**ECCV**)*, 2026.
13. Y. Luo, Y. Wang, X. Zou, S. Tang, X. Luo, Z. Zhou, L. Xiong, Y. Peng, C. Yang, N. Wang, H. Song, G. Zou, J. Shi, X. Zi, M. Gao, N. Jia, P. Yang, F. Yang, Z. Zheng, P. Wu, W. Dong, P. Dong, S. Guo, H. Han, S. Zhu, J. Chen, J. Luo, W. Zhai, Y. Xu, J. Chen, Y. Fan, L. Qu, X. Chen, J. Zhuang, H. Chen, C. Yu, X. Zhang, Q. Liu, F. Zhou, S. Zhang, **W. Luo**, X. Yao, Z. Zhang, ***Nature Communications***, 2026.
14. G. Liu, B. Yang, Y. Zhi, Z. Zhong, L. Ke, D. Deng, H. Gao, Y. Huang, K. Zhang, H. Fu, **W. Luo**, Beyond VLM-Based Rewards: Diffusion-Native Latent Reward Modeling, *International Conference on Machine Learning (**ICML**)*, 2026.
15. Z. Ye, Q. Liu, C. Wei, Y. Zhang, X. Wang, P. Wan, K. Gai, **W. Luo**, Visual-Aware CoT: Achieving High-Fidelity Visual Consistency in Unified Models, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.
16. F. Yang, X. Dong, X. Yu, **W. Luo**, W. Liu, K. Zhang, MRD: Multi-resolution Retrieval-Detection Fusion for High-Resolution Image Understanding, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.
17. M. Li, D. Wang, Z. Sun, J. Zhang, **W. Luo**, J. Guo, STiTch: Semantic Transition and Transportation in Collaboration for Training-Free Zero-Shot Composed Image Retrieval, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.
18. Y. Li, L. Liu, X. ZHANG, W. Xue, **W. Luo**, Y. Guo, Q. Tian, CogniEdit: Dense Gradient Flow Optimization for Fine-Grained Image Editing, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.
19. M. Li, P. Li, Z. Zhang, J. Lu, C. Zhao, W. Xue, Q. Liu, S. Peng, W. ZHANG, **W. Luo**, Y. Liu, Y. Guo, UniSH: Unifying Scene and Human Reconstruction in a Feed-Forward Pass, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.
20. L. Ke, H. Yin, G. Liu, Z. Lv, J. Guo, C. Li, **W. Luo**, Y. Yang, J. Lyu, FlowSteer:

Guiding Few-Step Image Synthesis with Authentic Trajectories, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.

21. Z. Ye, X. He, Q. Liu, Q. Wang, X. Wang, P. Wan, D. Zhang, K. Gai, Q. Chen, **W. Luo**, UNIC: Unified In-Context Video Editing, *International Conference on Learning Representations (**ICLR**)*, 2026.
22. Y. Li, Z. Wang, G. Li, W. Xue, **W. Luo**, Y. Guo, Pixel-Perfect Puppetry: Precision-Guided Enhancement for Face Image and Video Editing, *International Conference on Learning Representations (**ICLR**)*, 2026.
23. X. Liu, Y. He, L. Guo, X. Li, B. Jin, Y. Li, C. Chan, W. Xue, **W. Luo**, Q. Liu, Y. Guo, HiPrompt: Tuning-free Higher-Resolution Generation with Hierarchical MLLM Prompts, *International Journal of Computer Vision (**IJCV**)*, vol. 134, 2026.
24. Z. Kong, F. Gao, Y. Zhang, Z. Kang, X. Wei, X. Cai, G. Chen, **W. Luo**, Let Them Talk: Audio-Driven Multi-Person Conversational Video Generation, *Neural Information Processing Systems (**NeurIPS**)*, 2025.
25. Y. Cai, Z. Jiang, Y. Liu, C. Jiang, W. Xue, Y. Guo, **W. Luo**, Foundation Cures Personalization: Improving Personalized Models' Prompt Consistency via Hidden Foundation Knowledge, *Neural Information Processing Systems (**NeurIPS**)*, 2025.
26. Z. He, M. Yang, S. Yang, Y. Tang, T. Wang, K. Zhang, G. Chen, Y. Liu, J. Jiang, C. Guo, **W. Luo**, MaterialMVP: Illumination-Invariant Material Generation via Multi-view PBR Diffusion, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2025. (**Highlight**)
27. T. Wang, P. Xia, B. Li, P. Jiang, Z. Kong, K. Zhang, T. Lu, **W. Luo**, MOERL: When Mixture-of-Experts Meet Reinforcement Learning for Adverse Weather Image Restoration, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2025.
28. X. Chen, Y. Li, H. Shi, B. Hu, **W. Luo**, Y. Wang, M. Zhang, VideoVista-CulturalLingo: 360° Horizons-Bridging Cultures, Languages, and Domains in Video Comprehension, *The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2025.
29. Z. Kong, L. Li, Y. Zhang, F. Gao, S. Yang, T. Wang, K. Zhang, Z. Kang, X. Wei, G. Chen, **W. Luo**, DAM-VSR: Disentanglement of Appearance and Motion for Video Super-Resolution, ***ACM SIGGRAPH***, 2025.
30. P. Li, S. Ma, J. Chen, Y. Liu, C. Zhang, W. Xue, **W. Luo**, A. Sheffer, W. Wang, Y. Guo, CMD: Controllable Multiview Diffusion for 3D Editing and Progressive Generation, ***ACM SIGGRAPH***, 2025.
31. Z. Jin, Y. Qiu, K. Zhang, H. Li, **W. Luo**, MB-TaylorFormer V2: Improved Multi-branch Linear Transformer Expanded by Taylor Formula for Image Restoration, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 47, pp. 5990–6005, 2025.
32. Z. Ye, H. Huang, X. Wang, P. Wan, D. Zhang, **W. Luo**, StyleMaster: Stylize Your Video with Artistic Generation and Translation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2025.
33. P. Li, W. Zheng, Y. Liu, T. Yu, Y. Li, X. Qi, X. Chi, S. Xia, Y. Cao, W. Xue, **W. Luo**, Y. Guo, PSHuman: Photorealistic Single-image 3D Human Reconstruction using Cross-Scale Multiview Diffusion and Explicit Remeshing, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2025.
34. X. Mao, Z. Jiang, F. Wang, J. Zhang, H. Chen, M. Chi, Y. Wang, **W. Luo**, OSV: One Step is Enough for High-Quality Image to Video Generation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2025.
35. J. Xue, H. Wang, Q. Tian, Y. Ma, A. Wang, Z. Zhao, S. Min, W. Zhao, K. Zhang, H. Shum, W. Liu, M. Liu, **W. Luo**, Towards Multiple Character Image Animation Through Enhancing Implicit Decoupling, *International Conference on Learning Representations (**ICLR**)*, 2025.
36. X. Qi, Y. Wang, H. Zhang, J. Pan, W. Xue, S. Zhang, **W. Luo**, Q. Liu, Y. Guo, Co^{3}Gesture: Towards Coherent Concurrent Co-speech 3D Gesture Generation

with Interactive Diffusion, *International Conference on Learning Representations (**ICLR**)*, 2025.

37. Y. Li, S. Jiang, B. Hu, L. Wang, W. Zhong, **W. Luo**, L. Ma, M. Zhang, Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 47, pp. 3424-3439, 2025.
38. P. Li, Y. Liu, X. Long, F. Zhang, C. Lin, M. Li, X. Qi, S. Zhang, **W. Luo**, P. Tan, W. Wang, Q. Liu, Y. Guo, Era3D: High-Resolution Multiview Diffusion using Efficient Row-wise Attention, *Neural Information Processing Systems (**NeurIPS**)*, 2024.
39. L. Li, P. Dong, Z. Tang, X. Liu, Q. Wang, **W. Luo**, W. Xue, Q. Liu, X. Chu, Y. Guo, Discovering Sparsity Allocation for Layer-wise Pruning of Large Language Models, *Neural Information Processing Systems (**NeurIPS**)*, 2024.
40. T. Zhou, Q. Ye, **W. Luo**, H. Ran, Z. Shi, J. Chen, APPTracker+: Displacement Uncertainty for Occlusion Handling in Low-Frame-Rate Multiple Object Tracking, *International Journal of Computer Vision (**IJCV**)*, vol. 133, pp. 2044–2069, 2025.
41. J. Tan, H. Park, Y. Zhang, T. Wang, K. Zhang, X. Kong, P. Dai, Z. Liu, **W. Luo**, Blind Face Video Restoration with Temporal Consistent Generative Prior and Degradation-Aware Prompt, *The 32rd ACM International Conference on Multimedia (**ACM MM**)*, 2024.
42. Z. Kong, Y. Zhang, T. Yang, T. Wang, K. Zhang, B. Wu, G. Chen, W. Liu, **W. Luo**, OMG: Occlusion-friendly Personalized Multi-concept Generation In Diffusion Models, *European Conference on Computer Vision (**ECCV**)*, 2024.
43. B. Tang, K. Zhang, **W. Luo**, W. Liu, H. Li, Prompting Future Driven Diffusion Model for Hand Motion Prediction, *European Conference on Computer Vision (**ECCV**)*, 2024.
44. L. Li, H. Sun, S. Li, P. Dong, **W. Luo**, W. Xue, Q. Liu, Y. Guo, Auto-GAS: Automated Proxy Discovery for Training-free Generative Architecture Search, *European Conference on Computer Vision (**ECCV**)*, 2024.
45. L. Li, Z. Wei, P. Dong, **W. Luo**, W. Xue, Q. Liu, Y. Guo, AttnZero: Efficient Attention Discovery for Vision Transformers, *European Conference on Computer Vision (**ECCV**)*, 2024.
46. L. Li, Y. Bao, P. Dong, C. Yang, A. Li, **W. Luo**, Q. Liu, W. Xue, Y. Guo, DetKDS: Knowledge Distillation Search for Object Detectors, *International Conference on Machine Learning (**ICML**)*, 2024.
47. T. Wang, K. Zhang, Z. Shao, **W. Luo**, B. Stenger, T. Lu, T. Kim, W. Liu, H. Li, GridFormer: Residual Dense Transformer with Grid Structure for Image Restoration in Adverse Weather Conditions, *International Journal of Computer Vision (**IJCV**)*, vol. 132, pp. 4541–4563, 2024.
48. X. Qi, J. Pan, P. Li, R. Yuan, X. Chi, M. Li, **W. Luo**, W. Xue, S. Zhang, Q. Liu, Y. Guo, Weakly-Supervised Emotion Transition Learning for Diverse 3D Co-speech Gesture Generation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2024.
49. Y. Shao, S. He, Q. Ye, Y. Feng, **W. Luo**, J. Chen, Context-Aware Integration of Language and Visual References for Natural Language Tracking, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2024.
50. Y. Gao, W. Zhang, **W. Luo**, L. Ma, J. Yu, G. Xia, J. Ma, Aux-NAS: Exploiting Auxiliary Labels with Negligibly Extra Inference Cost, *International Conference on Learning Representations (**ICLR**)*, 2024.
51. W. Wang, C. Du, T. Wang, K. Zhang, **W. Luo**, L. Ma, W. Liu, X. Cao, Punctuation-level Attack: Single-shot and Single Punctuation Can Fool Text Models, *Neural Information Processing Systems (**NeurIPS**)*, 2023.
52. T. Zhou, Q. Ye, **W. Luo**, K. Zhang, Z. Shi, J. Chen, FnF Attack: Adversarial Attack against Multiple Object Trackers by Inducing False Negatives and False Positives, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2023.

53. P. Cheng, L. Lin, J. Lyu, Y. Huang, **W. Luo**, X. Tang, PRIOR: Prototype Representation Joint Learning from Medical Images and Reports, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2023.
54. Y. Qiu, K. Zhang, C. Wang, **W. Luo**, H. Li, Z. Jin, MB-TaylorFormer: Multi-branch Efficient Transformer Expanded by Taylor Formula for Image Dehazing, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2023.
55. S. Wang, C. Nguyen, J. Liu, K. Zhang, **W. Luo**, Y. Zhang, S. Muthu, F. Maken, H. Li, Homography Guided Temporal Fusion for Road Line and Marking Segmentation, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2023.
56. Z. Song, Z. Zhang, K. Zhang, **W. Luo**, Z. Fan, W. Ren and J. Lu, Robust Single Image Reflection Removal Against Adversarial Attacks, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2023.
57. T. Wang, K. Zhang, T. Shen, **W. Luo**, B. Stenger, T. Lu, Ultra-High-Definition Low-Light Image Enhancement: A Benchmark and Transformer-Based Method, *Proc. of the Association for the Advancement of Artificial Intelligence (**AAAI**)*, 2023. (**Oral Presentation**)
58. T. Zhou, **W. Luo**, Z. Shi, J. Chen, Q. Ye, APPTracker: Improving Tracking Multiple Objects in Low-Frame-Rate Videos, *The 30th ACM International Conference on Multimedia (**ACM MM**)*, 2022.
59. K. Zhang, D. Li, **W. Luo**, J. Liu, J. Deng, W. Liu, S. Zafeiriou, EDFace-Celeb-1M: Benchmarking Face Hallucination with a Million-scale Dataset, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 45, pp. 3968-3978, 2023.
60. K. Zhang, W. Ren, **W. Luo**, W. Lai, B. Stenger, M. Yang, H. Li, Deep Image Deblurring: A Survey, *International Journal of Computer Vision (**IJCV**)*, vol. 130, pp. 2103-2130, 2022.
61. K. Zhang, **W. Luo**, Y. Yu, W. Ren, F. Zhao, C. Li, L. Ma, W. Liu, H. Li, Beyond Monocular Deraining: Parallel Stereo Deraining Network Via Semantic Prior, *International Journal of Computer Vision (**IJCV**)*, vol. 130, pp. 1754-1769, 2022.
62. Y. Wang, Guo Pu, **W. Luo**, Y. Wang, P. Xiong, H. Kang, Z. Lian, Aesthetic Text Logo Synthesis via Content-aware Layout Inferring, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2022.
63. K. Zhang, D. Li, **W. Luo**, W. Ren, W. Liu, Enhanced Spatio-Temporal Interaction Learning for Video Deraining: A Faster and Better Framework, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 45, pp. 1287-1293, 2023.
64. K. Zhang, D. Li, **W. Luo**, W. Ren, B. Stenger, W. Liu, H. Li, M. Yang, Benchmarking Ultra-High-Definition Image Super-resolution, *Proc. of International Conference on Computer Vision (**ICCV**)*, 2021.
65. W. Niu, K. Zhang, **W. Luo**, Y. Zhong, Blind Motion Deblurring Super-Resolution: When Dynamic Spatio-Temporal Learning Meets Static Image. *IEEE Trans. on Image Processing (**TIP**)*, vol. 30, pp. 7101-7111, 2021.
66. K. Zhang, D. Li, **W. Luo**, W. Ren, Dual Attention-in-Attention Model for Joint Rain Streak and Raindrop Removal. *IEEE Trans. on Image Processing (**TIP**)*, vol. 30, pp. 7608-7619, 2021.
67. K. Zhang, R. Li, Y. Yu, **W. Luo**, C. Li, Deep Dense Multi-scale Network for Snow Removal Using Semantic and Geometric Priors. *IEEE Trans. on Image Processing (**TIP**)*, vol. 30, pp. 7419-7431, 2021.
68. F. Zhong, P. Sun, **W. Luo**, T. Yan, Y. Wang, Towards Distraction-Robust Active Visual Tracking, *International Conference on Machine Learning (**ICML**)*, 2021.
69. X. Zhang, R. Jiang, T. Wang and **W. Luo**, Single Image Dehazing via Dual-Path Recurrent Network, *IEEE Trans. on Image Processing (**TIP**)*, vol. 30, pp. 5211-5222, 2021.
70. W. Liu, Z. Piao, Z. Tu, **W. Luo**, L. Ma, and S. Gao, Liquid Warping GAN with Attention: A Unified Framework for Human Image Synthesis, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 44, pp. 5114-5132, 2022.

71. **W. Luo**, J. Xing, A. Milan, X. Zhang, W. Liu, and T-K. Kim, Multiple Object Tracking: A Literature Review, ***Artificial Intelligence***, vol. 293, pp. 103448, 2021.
72. Z. Ren, **W. Luo**, J. Yan, X. Yang, A. Yuille, H. Zha, STFlow: Self-Taught Optical Flow Estimation Using Pseudo Labels, *IEEE Trans. on Image Processing (**TIP**)*, vol. 29, pp. 9113-9124, 2020.
73. T. Liu, **W. Luo**, L. Ma, J. Huang, T. Stathaki, T. Dai, Coupled Network for Robust Pedestrian Detection with Gated Multi-Layer Feature Extraction and Deformable Occlusion Handling, *IEEE Trans. on Image Processing (**TIP**)*, vol. 30, pp. 754-766, 2021.
74. K. Zhang, **W. Luo**, B. Stenger, W. Ren, L. Ma, H. Li, Every Moment Matters: Detail-Aware Networks to Bring a Blurry Image Alive, *The 28th ACM International Conference on Multimedia (**ACM MM**)*, 2020.
75. K. Zhang, **W. Luo**, W. Ren, J. Wang, F. Zhao, L. Ma, H. Li, Beyond Monocular Deraining: Stereo Image Deraining via Semantic Understanding, *European Conference on Computer Vision (**ECCV**)*, UK, 2020.
76. K. Zhang, **W. Luo**, Y. Zhong, L. Ma, B. Stenger, W. Liu, H. Li, Deblurring by Realistic Blurring, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2020. (**Oral Presentation**)
77. W. Xiong, Y. He, Y. Zhang, **W. Luo**, L. Ma, J. Luo, Fine-grained Image-to-Image Transformation towards Visual Recognition, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2020.
78. F. Zhong, P. Sun, **W. Luo**, T. Yan, Y. Wang, AD-VAT+: An Asymmetric Dueling mechanism for learning and understanding Visual Active Tracking, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 43, pp. 1467-1482, 2021.
79. Z. Liu, **W. Luo**, B. Wu, X. Yang, W. Liu, K-T. Cheng, Bi-Real Net: Binarizing Deep Network Towards Real-Network Performance, *International Journal of Computer Vision (**IJCV**)*, vol. 128, pp. 202-219, 2020.
80. W. Liu, Z. Piao, J. Min, **W. Luo**, L. Ma, S. Gao, Liquid Warping GAN: A Unified Framework for Human Motion Imitation, Appearance Transfer and Novel View Synthesis, *Proc. of International Conference on Computer Vision (**ICCV**)*, Korea, 2019.
81. Z. Chen, L. Ma, **W. Luo**, K.-Y. K. Wong, Weakly-Supervised Spatio-Temporally Grounding Natural Sentence in Video, *The 57th Annual Meeting of the Association for Computational Linguistics (**ACL**)*, Italy, 2019. (**Oral Presentation**)
82. X. Yang, **W. Luo**, L. Bao, Y. Gao, D. Gong, S. Zheng, Z. Li, W. Liu, Face Anti-Spoofing: Model Matters, So Does Data, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2019.
83. K. Zhang, **W. Luo**, L. Ma, W. Liu, H. Li, Learning Joint Gait Representation via Quintuplet Loss Minimization, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2019. (**Oral Presentation**)
84. J. Wan, **W. Luo**, B. Wu, A. Chan, W Liu, Residual Regression with Semantic Prior for Crowd Counting, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2019.
85. K. Tang, H. Zhang, B. Wu, **W. Luo**, W. Liu, Learning to Compose Dynamic Tree Structures for Visual Contexts, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, USA, 2019. (**Oral Presentation, Best Paper Finalist**)
86. **W. Luo**, P. Sun, F. Zhong, W. Liu, T. Zhang and Y. Wang, End-to-end Active Object Tracking and Its Real-world Deployment via Reinforcement Learning, *IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**)*, vol. 42, pp. 1317-1332, 2020.
87. F. Zhong, P. Sun, **W. Luo**, T. Yan, Y. Wang, AD-VAT: An Asymmetric Dueling mechanism for learning Visual Active Tracking, *International Conference on Learning Representations (**ICLR**)*, 2019.

88. K. Zhang, **W. Luo**, L. Ma, H. Li, Cousin Network Guided Sketch Recognition via Latent Attribute Warehouse, *Proc. of the Association for the Advancement of Artificial Intelligence (**AAAI**)*, 2019. (**Spotlight Presentation**)
89. **W. Luo**, B. Stenger, X. Zhao, T-K. Kim, Trajectories as Topics: Multi-Object Tracking by Topic Discovery, *IEEE Trans. on Image Processing (**TIP**)*, vol. 28, no. 1, pp. 240-252, 2019.
90. K. Zhang, **W. Luo**, Y. Zhong, L. Ma, W. Liu, H. Li, Adversarial Spatio-Temporal Learning for Video Deblurring, *IEEE Trans. on Image Processing (**TIP**)*, vol. 28, no. 1, pp. 291-301, 2019.
91. Z. Liu, B. Wu, **W. Luo**, X. Yang, W. Liu, K-T. Cheng, Bi-Real Net: Enhancing the Performance of 1-bit CNNs with Improved Representational Capability and Advanced Training Algorithm, *European Conference on Computer Vision (**ECCV**)*, Germany, 2018.
92. **W. Luo**, P. Sun, F. Zhong, W. Liu, T. Zhang and Y. Wang, End-to-end Active Object Tracking via Reinforcement Learning, *International Conference on Machine Learning (**ICML**)*, Sweden, 2018.
93. W. Xiong, **W. Luo**, L. Ma, W. Liu and J. Luo, Learning to Generate Time-Lapse Videos Using Multi-Stage Dynamic Generative Adversarial Networks. *IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2018.
94. H. Huang, H. Wang, **W. Luo**, L. Ma, W. Jiang, X. Zhu, Z. Li, W. Liu, Real-Time Neural Style Transfer for Videos. *IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2017.
95. **W. Luo**, B. Stenger, X. Zhao, T-K. Kim. Automatic Topic Discovery for Multi-object Tracking. *Proc. of the Association for the Advancement of Artificial Intelligence (**AAAI**)*, 2015. (**Oral Presentation**)
96. **W. Luo**, T-K. Kim, B. Stenger, X. Zhao, R. Cipolla. Bi-label Propagation for Generic Multiple Object Tracking. *IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2014.
97. X. Zhao, T-K. Kim, **W. Luo**. Unified Face Analysis by Iterative Multi-Output Random Forests. *IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2014.
98. W. Hu, X. Zhou, W. Li, **W. Luo**, X. Zhang, S. Maybank. Active Contour-Based Visual Tracking by Integrating Colors, Shapes and Motions. *IEEE Trans. on Image Processing (**TIP**)*, vol. 22, no. 5, pp. 1778-1792, 2013.
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Courses

- AMCC5210, Practice and Techniques for Visual Creatives, 2025/26 Fall, 2026/27 Fall (PG)
- ARIN 5201, Machine Learning for Computer Vision, 2025/26 Fall, 2026/27 Fall (PG)
- MAIE 5421, Computer Vision, 2025/26 Fall, 2026/27 Fall (PG)
- ARIN 5302, Medical Image Analysis, 2025/26 Spring (PG)
- EMIA4110, Practical Machine Learning, 2024-25 Fall, 2024/25 Spring, 2025/26 Spring (UG)
- EMIA4991, Interdisciplinary Capstone Project, 2024/25 Spring (UG)
- EMIA6950C/F, Independent Study, 2024/25 Spring, 2025/26 Fall, 2025/26 Spring (PG)
- AMCC6900C, Independent Study in Arts and Machine Creativity, 2025/26 Spring (PG)
- EMIA4900B, Independent Study, 2025/26 Spring, 2025/26 Summer (UG)
- EMIA2010B, Cross-disciplinary Seminar: Digital Media and Creative Arts, 2024/25

Fall (UG)

Professional Service

- **Senior Area Editor**: IEEE Signal Processing Letters (2025 - present)
- **Associate Editor**: IEEE Transactions on Image Processing (2025 - present), Neurocomputing (2024 - present), IET Computer Vision (2024 - present)
- **Guest Editor**: Pattern Recognition (Jun 2026 -), Computer Vision and Image Understanding (Sep 2020 - Sep 2021)
- **Senior Area Chair**: NeurIPS 2026
- **Area Chair**: ICLR (2027, 2026), NeurIPS 2025, ACM MM (2026, 2025), ICML (2026, 2025), IJCAI 2025, IJCNN 2025, BMVC (2025, 2024)
- **Conference Organization/Service**: SIGGRAPH Asia 2025 (Conflict-of-Interest Coordinator), CVM 2025 (Workshop Chair), AsiaGraphics (Vice Secretary, 2025 - present)
- **Senior Program Committee**: AAAI (2027, 2026, 2025, 2024, 2022), IJCAI (2026, 2024, 2023, 2021)
- **Workshop Organizer**: Consistency in Video Generative Models: from Clip to Wild, in conjuction with AAAI 2026
- **Journal Reviewer**: TPAMI, TIP, CSUR, TNNLS, TMM, TCSVT, PR
- **Conference Reviewer**: ICML, NeurIPS, CVPR, ACL, ECCV, ICCV, BMVC, AAAI, IJCAI, ACM MM, WACV

University and School Services

- Member of University UG Common Core Course Review Panel (CRP), 2026 - present
- Interdisciplinary PG Studies Committee, Academy of Interdisciplinary Studies, 2025 - present
- Search Committee, for the Head of the Div. Arts and Machine Creativity, 2024

Departmental Services

- Chair of Undergraduate Program Committee, Div. of Arts and Machine Creativity, 2026 - present
- PG Programs Coordinator, Div. of Arts and Machine Creativity, 2025 - present
- Chair of PG Committee, Div. of Arts and Machine Creativity, 2025 - present
- Search and Appointment Committee (SAC), Div. of Arts and Machine Creativity, 2024 - present
- 3-in-1 Planning Committee (Research Matters), Div. of Arts and Machine Creativity, 2025 - present
- Deputy Critical Incident Coordinator, Div. of Arts and Machine Creativity, 2025 - present
- Search and Appointment Committee (SAC), Div. of Emerging Interdisciplinary Areas, 2024 - present
- PG Committee, Div. of Emerging Interdisciplinary Areas, 2024 - 2025

Talks

- Invited talk in the Young Scientists Forum in Lianhua Mountain Artificial Intelligence Conference 2026, Jul 2026
- Invited talk in The Forum on Low-Quality Visual Processing and Quality Assessment in Chinese Congress on Image and Graphics 2026, May 2026
- Invited talk in The Asiagraphics Workshop on Intelligent Graphics in SIGGRAPH Asia 2025, Dec 2025
- Invited talk in Tsinghua University, hosted by Prof. Yujiu Yang, Sep 2025
- Invited talk in the Institute of Automation, Chinese Academy of Sciences, hosted by Prof. Jin Gao, Aug 2025.
- Invited talk in The Forum on Multimodal Content Understanding and Generation in China Spatial Intelligence Conference 2025, Jul 2025

- Invited talk in the Institute of Automation, Chinese Academy of Sciences, hosted by Prof. Jin Gao, Aug 2025.
- Shenzhen University, hosted by Prof. Yanshan Li, Mar 2025.
- Shenzhen University, hosted by Prof. Xu Wang, Dec 2024.
- Zhejiang University, hosted by Prof. Qi Ye, Oct 2024.
- Southern University of Science and Technology, hosted by Prof. Feng Zheng, Jul 2024.
- Wuhan University, hosted by Prof. Guisong Xia, Apr 2024
- Zhejiang University, hosted by Dr. Qi Ye, Jun 2022.
- Beijing Institute of Technology, hosted by Prof. Changsheng Li, Jan 2022.
- Southern University of Science and Technology, hosted by Prof. Xiaoying Tang, Sep 2020.
- The Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, hosted by Prof. Yun Zhang, Sep 2018.
- The Institute of Automation, Chinese Academy of Sciences, hosted by Prof. Weiming Hu, Aug 2018.

Patents

- **W. Luo**, L. Bao, Y. Gao, W. Liu, Method and apparatus for generating negative sample of face recognition, and computer device, US 11302118 B2
- **W. Luo**, Y. Wang, W. Liu, Living body detection method and apparatus, electronic device, storage medium, and related system to which living body detection method is applied, US 2021/0073516 A1
- **W. Luo**, L. Ma, W. Liu, Neural network model training method and device, and time-lapse photography video generating method and device, US 2020/0293833 A1
- **W. Luo**, Target Tracking Method And Apparatus, Electronic Device, And Storage Medium, US 11145069 B2
- K. Zhang, **W. Luo**, L. Ma, W. Liu, Video Deblurring Method And Apparatus, Storage Medium, And Electronic Apparatus, US 2020/0372618 A1
- K. Zhang, **W. Luo**, L. Ma, W. Liu, Media processing method, related apparatus, and storage medium, US 11335127 B2
- K. Zhang, **W. Luo**, L. Ma, W. Liu, Image recognition method and apparatus, terminal, and storage medium, US2020/0302180 A1
- K. Zhang, **W. Luo**, H. Li, Image processing method and apparatus, computer device, and storage medium, US 2021/0256663 A1
- K. Zhang, **W. Luo**, L. Ma and W. Liu, Microexpression-based image recognition method and apparatus, and related device. US20210174072A1
- Z. Chen, L. Ma, **W. Luo**, W Liu, Video sequence selection method, computer device, and storage medium, US 2021/0224601 A1
- Z. Chen, L. Ma, **W. Luo**, W Liu, Video clip positioning method and apparatus, computer device, and storage medium, US 2021/0349940 A1
- H. Huang, H. Wang, **W. Luo**, L. Ma, W. Jiang, X. Zhu, Z. Li, W. Liu, Method And Apparatus For Training Neural Network Model Used For Image Processing, And Storage Medium, US 2021/0182616 A1

Thesis

W. Luo, Generic Multiple Object Tracking, Dept. of Electrical and Electronic Engineering, Imperial College London, 2016.